

SQL PROJECT: BANKING TRANSACTION ANALYSIS

Arianty Siahaan-
Data Analyst

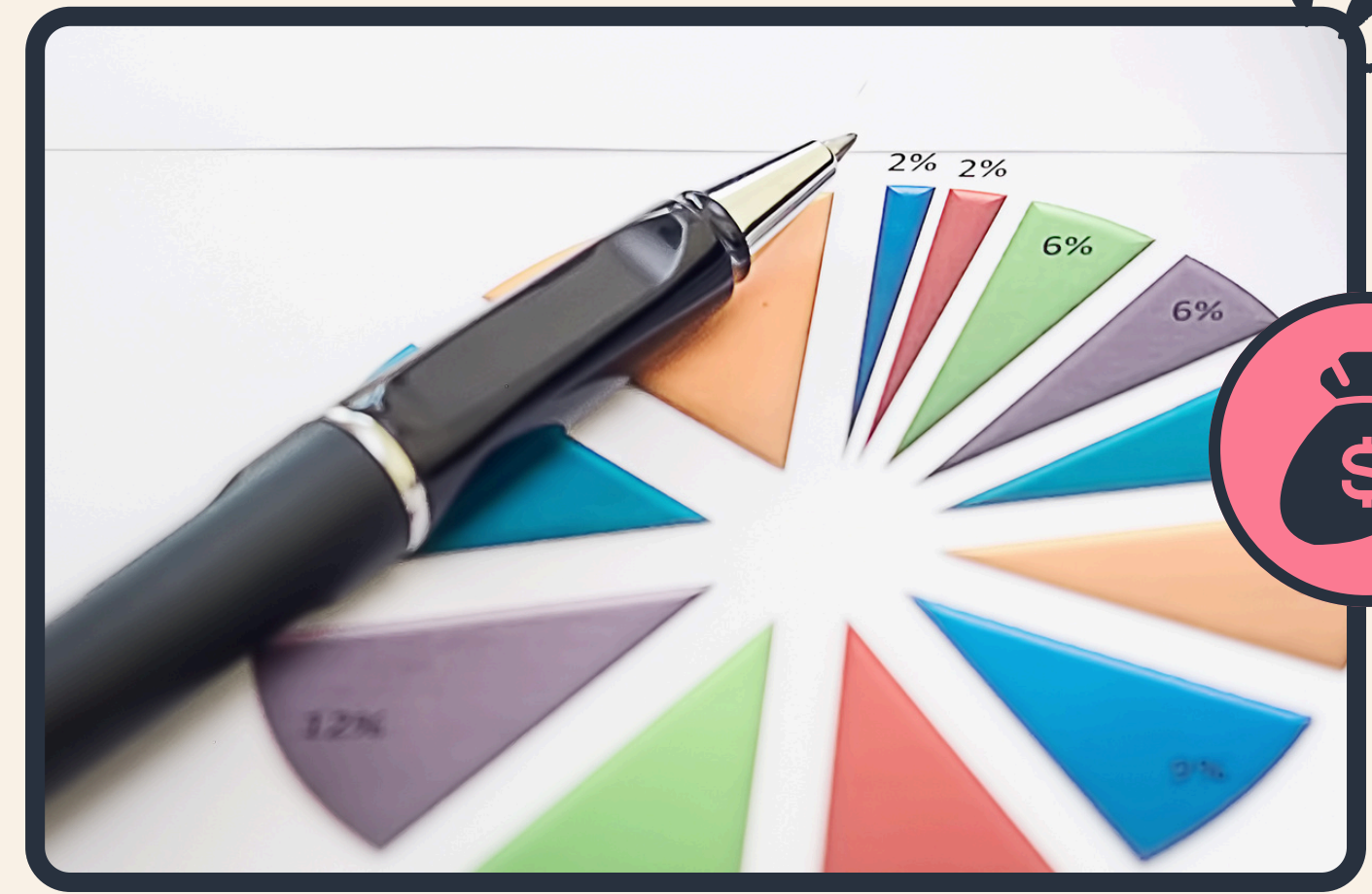


Table account

123 account_id ▼	123 customer_id ▼	A-Z account_type ▼	123 balance ▼	A-Z open_date ▼
101	1	Saving	15,000,000	2019-01-10
102	2	Saving	8,000,000	2018-05-21
103	3	Premium	45,000,000	2021-07-01
104	4	Premium	70,000,000	2017-11-20
105	5	Saving	5,000,000	2022-02-12

Table customer

123 customer_id ▼	A-Z customer_name ▼	A-Z gender ▼	123 birth_year ▼	A-Z city ▼
1	Andi	M	1,988	Jakarta
2	Budi	M	1,982	Bandung
3	Citra	F	1,995	Surabaya
4	Dewi	F	1,978	Jakarta
5	Eka	M	1,992	Medan

Table transaksi

123 transaction_id ▼	123 account_id ▼	A-Z transaction_date ▼	123 amount ▼	A-Z transaction_type ▼
1,001	101	2024-01-05	2,000,000	debit
1,002	101	2024-01-10	5,000,000	debit
1,003	102	2024-01-08	1,000,000	debit
1,004	103	2024-01-15	15,000,000	debit
1,005	104	2024-01-18	30,000,000	debit
1,006	104	2024-01-20	10,000,000	debit
1,007	103	2024-01-22	5,000,000	credit

QUESTION 1



Display all
customers who
live in Jakarta





ANSWER



```
select customer_id, customer_name, city  
from customers  
where city = 'Jakarta';
```

Result

	123 customer_id ▼	A-Z customer_name ▼	A-Z city ▼
	1	Andi	Jakarta
	4	Dewi	Jakarta

QUESTION 2



Calculate the
number of
accounts by
account type



ANSWER



```
select account_type, count (*) as total_account  
from accounts a  
group by account_type;
```

Result

	A-Z account_type ▼	123 total_account ▼
1	Saving	3
2	Premium	2

QUESTION 3



Display the
total debit
transaction
amount



ANSWER



```
SELECT  
  transaction_type,  
  SUM(amount) AS total_transaction  
FROM transactions  
WHERE transaction_type = 'debit'  
GROUP BY transaction_type;
```

Result

	A-Z transaction_type ▼	123 total_transaction ▼
1	debit	63,000,000

QUESTION 4



Calculate the
total debit
transactions
per customer

ANSWER

```
select
  a.customer_id,
  SUM(t.amount) AS total_transaction
from transactions t
join accounts a
  on a.account_id = t.account_id
where t.transaction_type = 'debit'
group by a.customer_id;
```

Result

123 customer_id ▼	123 total_transaction ▼
1	7,000,000
2	1,000,000
3	15,000,000
4	40,000,000

QUESTION 5



- Display:
- customer_name
 - city
 - total debit transactions

ANSWER

```
with gab_cus_acc as (  
  select  
    a.account_id,  
    c.customer_name,  
    c.city  
  from accounts a  
  join customers c  
    on a.customer_id = c.customer_id  
)  
select gca.customer_name, gca.city, sum(t.amount) as total_transaction  
from gab_cus_acc gca  
join transactions t on gca.account_id = t.account_id  
where t.transaction_type = 'debit'  
group by gca.customer_name, gca.city;
```

Result

	A-Z customer_name ▼	A-Z city ▼	123 total_transaction ▼
1	Andi	Jakarta	7,000,000
2	Budi	Bandung	1,000,000
3	Citra	Surabaya	15,000,000
4	Dewi	Jakarta	40,000,000

QUESTION 6



Calculate the
average account
balance by account
type





ANSWER



```
select  
  account_type,  
  avg(balance) as avg_saldo  
from accounts  
group by account_type;
```

Result

A-Z account_type ▼	123 avg_saldo ▼
Saving	9,333,333.333333333333
Premium	57,500,000

QUESTION 7



Display the top 3 customers with the highest total debit transactions



ANSWER



```
with merg_acc_cust as (select a.customer_id, a.account_id, c.customer_name
                           from accounts a
                           join customers c on a.customer_id=c.customer_id)
select mac.customer_id, mac.customer_name, sum(t.amount) as total_transaction
from transactions t
join merg_acc_cust mac on t.account_id=mac.account_id
where t.transaction_type='debit'
group by mac.customer_id, mac.customer_name
order by total_transaction desc
limit 3;
```

Result

123 customer_id ▼	A-Z customer_name ▼	123 total_transaction ▼
4	Dewi	40,000,000
3	Citra	15,000,000
1	Andi	7,000,000

QUESTION 8



Find customers whose total debit transactions are greater than their account balance

ANSWER



```
with customer_debit as (  
  select  
    a.customer_id,  
    sum(t.amount) as total_debit  
  from accounts a  
  join transactions t on a.account_id = t.account_id  
  where t.transaction_type = 'debit'  
  group by a.customer_id  
)  
customer_balance as (  
  select  
    customer_id,  
    sum(balance) as total_balance  
  from accounts  
  group by customer_id  
)  
select  
  cd.customer_id,  
  cd.total_debit,  
  cb.total_balance  
from customer_debit cd  
join customer_balance cb  
  on cd.customer_id = cb.customer_id  
where cd.total_debit > cb.total_balance;
```

Result

	123 customer_id ▼	123 total_debit ▼	123 total_balance ▼

QUESTION 9



Calculate the total transactions per city



ANSWER

```
with account_city as (  
  select  
    a.account_id,  
    c.city  
  from accounts a  
  join customers c  
    on a.customer_id = c.customer_id  
)  
  
select  
  ac.city,  
  sum(t.amount) as total_transaction_amount  
from transactions t  
join account_city ac  
  on t.account_id = ac.account_id  
group by ac.city  
order by total_transaction_amount desc;
```

Result

A-Z city ▼	123 total_transaction_amount ▼
Jakarta	47,000,000
Surabaya	20,000,000
Bandung	1,000,000

QUESTION 10



1. Display customers who:
 - have an account type of Premium
 - have more than one debit transaction



ANSWER

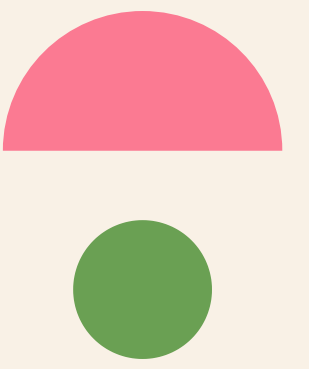
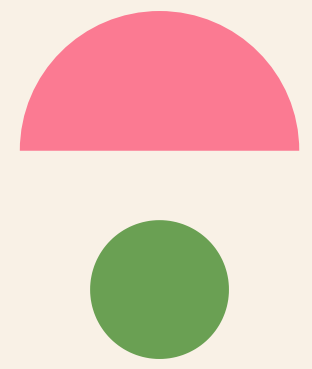


```
with merg_acc_type as
    (select a.account_id, a.account_type
     from accounts a
     join transactions t on a.account_id = t.account_id
     where a.account_type='Premium'
     and t.transaction_type='debit')

select mat.account_id, mat.account_type, count(*) as total_debit
from merg_acc_type mat
group by mat.account_id, mat.account_type
having count(*)>1;
```

Result

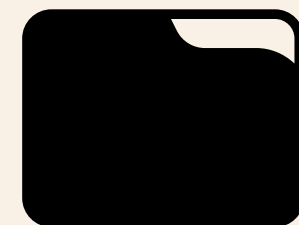
123 account_id ▼	A-Z account_type ▼	123 total_debit ▼	
104	Premium	2	



THANK YOU



Linkedin:
Arianty Siahaan



Portfolio:
<https://linktr.ee/AriantySiahaan>